

Entropy Collapse Under a Small Sample Budget in Physics-in-the-Loop MAPPO: A Diagnosis That Did Not Reproduce

Harsh Pandhe¹

¹Independent Research , harshpandhehome@gmail.com

October 2026

Abstract

Training multi-robot policies against a general-purpose, real-time simulator (Gazebo) limits a single experiment to thousands rather than millions of environment steps. In an early run of a centralized-critic MAPPO policy for three-robot area coverage (15 iterations, 4,500 steps, RLlib default GAE $\lambda = 1.0$ and entropy coefficient 0.0), we recorded policy entropy falling to -0.53 and episode returns between -227 and $+3$, and attributed this to the default hyperparameters. We proposed a correction (smaller λ , a nonzero entropy bonus, a larger batch, shorter episodes and a rebalanced reward). The raw logs of that run were not kept, and the attribution was never tested against a control. Here we test it: six independent training runs, three per configuration, with per-iteration metrics logged. We did not reproduce the collapse. Under the original defaults, entropy at iteration 15 was 1.75, 1.50 and 1.87 (minimum 0.79 in one run, which then recovered), nowhere near -0.53 . Under the corrected configuration it was 1.95, 2.12 and 2.48, higher in all three runs (entropy rose by 0.36 to 0.90 over training, against -0.45 to $+0.48$ for the original), but with three runs per arm the smallest attainable two-sided permutation p is 0.10, which is also the value obtained. Per-episode return spread did not differ between arms (72 vs. 67), and mean returns were negative throughout and did not clearly improve in either arm. The comparison changes five things at once and logs no task metric, so it cannot say which change matters or whether the correction improves coverage. We conclude that the original diagnosis, built from one unreplicated run, is not supported, that run-to-run variability in entropy is large enough for a single run to mislead, and that the corrected configuration raises entropy modestly without our having shown a benefit.

Keywords: multi-agent reinforcement learning, MAPPO, entropy collapse, sample budget, reproducibility, negative results

1 Introduction

Sample-efficiency work in deep reinforcement learning usually assumes that environment steps are cheap or parallelizable. General-purpose robotics simulators used for sensor-level fidelity (LiDAR ray casting, collision geometry, ROS integration) are often run as a single real-time instance, which makes a budget of thousands of steps per experiment a practical ceiling rather than a choice. Under such budgets a batch contains only a few complete episodes, and the usual intuitions about advantage-estimation variance and exploration pressure are worth examining.

This paper began as a diagnosis. After one early training run showed what looked like entropy collapse, we traced it, by reasoning about the configuration, to RLlib’s default $\lambda = 1.0$ and entropy coefficient 0.0, proposed a correction, and wrote it up as a mechanistic case study. It is now a report on testing that diagnosis. We ran the control we should have run first, and the result does not support the original account. We report it because the failure to reproduce is informative: it shows how a causal story assembled from a single unreplicated run can look convincing and be wrong, and it gives a replicated baseline for entropy trajectories in this setting. The claim is narrow. We do not claim that default hyperparameters are safe in small-batch multi-agent training in general, only that in this system, with three runs per arm, we did not see the collapse we reported.

2 Related Work

Yu et al. [6] establish MAPPO with a centralized critic as a strong baseline for cooperative multi-agent games; their results are reported with far more environment steps than are available here. Work aimed at closing the sample-budget gap includes differentiable surrogate models alongside a non-differentiable simulator [7] and world-model-based learning without RL [5], which require a learned or differentiable dynamics model that we do not assume. Masked reconstruction as an auxiliary task improves sample efficiency in value-based MARL [2], and physics-informed structure has been used for distributed multi-robot learning [4]; both are complementary to the hyperparameter-level question studied here. Henderson et al. [1] show that results in deep RL depend heavily on random seeds, implementation details and the number of runs, which is the concern our replication illustrates.

A companion study [3] evaluates the corrected configuration’s downstream coverage performance against a classical planner.

3 System and the Original Diagnosis

The system is a shared MAPPO policy for three TurtleBot3 robots performing cooperative area coverage in Gazebo Harmonic (ROS 2 Jazzy), trained with Ray RLlib. Each robot observes a 46-dimensional vector: 24 LiDAR ranges, goal distance and bearing, linear and angular velocity, and nine neighbor slots of (distance, bearing). Each neighbor pair is embedded by a two-layer ReLU network ($2 \rightarrow 16 \rightarrow 16$), masked and mean-pooled, concatenated with the 28 ego features, and passed through a two-layer Tanh actor ($44 \rightarrow 64 \rightarrow 64$). A centralized critic ($76 \rightarrow 64 \rightarrow 64 \rightarrow 1$, Tanh) takes the same 44 features plus a mean-pooled embedding ($48 \rightarrow 32 \rightarrow 32$, ReLU) of the other agents’ observation–action pairs. Training runs with `num_env_runners=0`, since Gazebo must run in the main process, so all rollouts are serial. Learning rate 10^{-4} , PPO clip 0.2, $\gamma = 0.99$, 5 epochs and minibatch size 64 are common to all runs.

Original observation. The first configuration left GAE λ and the entropy coefficient at RLlib’s defaults (1.0 and 0.0), used a train batch of 300 steps, 150-step episodes, a collision penalty of -20 and a coverage weight of 2.0. In one 15-iteration run (4,500 steps) we recorded final-iteration entropy of -0.53 and episode returns between -227 and $+3$. These figures come from our notes of that run; its logs were not archived, and we cannot re-derive them. The original account attributed the entropy to two mechanisms: a $\lambda = 1.0$ Monte-Carlo estimator is high-variance when a batch holds only a few episodes, and a zero entropy coefficient exerts no pressure against premature convergence. Neither mechanism was tested. We also stated that a batch held “three to four” episodes; in the runs reported below it held two.

Proposed correction. $\lambda = 0.95$; entropy coefficient 0.01; train batch 700 with 90-step episodes (more distinct episodes per iteration for similar wall-clock time, since episode-reset overhead is fixed); collision penalty -8 and coverage weight 4.0. We considered but did not pursue curriculum learning and behavior-cloning warm starts.

4 Controlled Test

Design. Two configurations, *original defaults* and *corrected*, each trained for 15 iterations, three independent runs per configuration, interleaved in time (original, corrected, original, ...) so that any drift in the simulator or host affects both. Everything not listed in Section 3

is identical, including code and world. The Gazebo seed was fixed and RLlib’s seed was not set, so runs differ through RLlib’s own randomness and simulator timing. Per-iteration policy entropy, losses, value-function explained variance and episode returns were logged. The unit of independence is the training run, so $n = 3$ per arm. The analysis plan was fixed before we looked at any run’s results: the primary outcome is entropy at iteration 15 (original lower than corrected); the secondary outcome is the change in entropy from iteration 1 to 15; return spread is descriptive only because return scales differ between arms. One change was made after the first run finished: we replaced the within-iteration range of returns, which grows with the number of episodes (two against eight per iteration), with the standard deviation of individual episode returns. Tests are exact two-sided permutation tests on the difference of arm means. With three runs per arm there are 20 permutations, so the smallest attainable p is 0.10.

What the comparison cannot separate. The two configurations differ in λ , entropy coefficient, batch size, episode length and reward scale, and they collect different numbers of steps (4,500 against 10,800, because sampling ends on whole episodes; the nominal 15×700 is 10,500). We therefore cannot attribute any difference to a single change. No coverage or other task metric was logged and the checkpoints were not retained, so we say nothing here about task performance.

Table 1: Six independent 15-iteration training runs. Entropy is the policy entropy logged by RLlib on each iteration’s training batch; Δ is iteration 15 minus iteration 1. Return columns are over the new episodes of each iteration (30 episodes per original run, about 120 per corrected run). Return scales differ between configurations.

Configuration	Run	Ent. it. 1	Ent. it. 15	Δ	Ent. min	Mean return	Return SD	Steps
Original defaults	1	1.61	1.75	+0.14	1.43	−59	66	4,500
Original defaults	2	1.95	1.50	−0.45	0.79	−62	67	4,500
Original defaults	3	1.39	1.87	+0.48	1.39	−66	84	4,500
Corrected config	1	1.59	1.95	+0.36	1.43	−43	61	10,800
Corrected config	2	1.59	2.12	+0.53	1.50	−51	69	10,800
Corrected config	3	1.58	2.48	+0.90	1.58	−56	71	10,800

4.1 Results

Entropy did not collapse under the original defaults. Entropy at iteration 15 was 1.75, 1.50 and 1.87 (mean 1.71). The lowest value in any original run was 0.79 (run 2, iteration 12), after which it rose to 1.50. No run approached the reported −0.53.

The corrected configuration gave higher entropy, not significantly. Entropy at iteration 15 was 1.95, 2.12 and 2.48 (mean 2.18); the difference in means is −0.475 (original

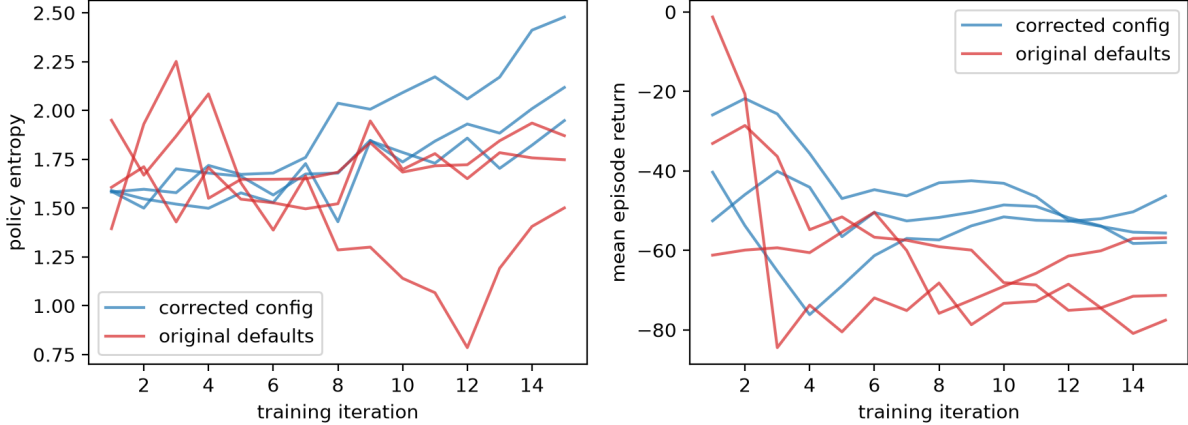


Figure 1: Policy entropy (left) and mean episode return per iteration (right) for the six runs, three per configuration. The original defaults (red) do not collapse; one run dips to 0.79 at iteration 12 and recovers.

minus corrected), exact $p = 0.10$, the smallest value attainable at $n = 3$. Entropy rose over training in all three corrected runs (+0.36, +0.53, +0.90) and was mixed in the original runs (+0.14, -0.45, +0.48); the difference in the change is -0.54, $p = 0.20$. Mean entropy over the 15 iterations differed by only -0.16 ($p = 0.20$). Part of the corrected arm’s higher entropy is mechanical, since an entropy bonus rewards it directly, so a difference of this kind does not by itself show better learning.

Return spread did not differ. The standard deviation of per-episode returns was 72 (original, pooled) against 67 (corrected); run-level values were 66, 67, 84 and 61, 69, 71 ($p = 0.6$). Individual episode returns reached about -350 in both arms, so the wide swings originally attributed to the defaults are not specific to them. Mean returns did not improve with training. Comparing the last five iterations with the first five, mean return was lower in four of the six runs, essentially unchanged in one (original run 2, -59 → -60) and higher in one (corrected run 3, -61 → -51). Averaged over all iterations the original runs gave -59, -62 and -66 and the corrected runs -43, -51 and -56, but the arms use different collision penalties and episode lengths, so we do not compare these directly.

5 Discussion

The original diagnosis does not survive a replicated test. The mechanisms it proposed remain plausible in principle, but the symptom they were meant to explain, entropy collapse under the defaults, did not appear in three attempts. Two explanations are consistent with this. The original run may have been an unrepresentative draw: the original arm’s entropy

trajectories are variable (final values 1.50 to 1.87, one transient dip to 0.79), so a single run can look like collapse or like nothing. Or the early run differed from these in some respect we did not record, since the code changed between them in ways not captured in the notes. We cannot tell these apart, and we do not claim the first.

The corrected configuration raises entropy modestly and, in this sample, consistently in direction, but the sample is too small to call it an effect, and we did not measure whether it improves the task. The companion study [3] trained the corrected configuration for 45 iterations and found it did not close the gap to a classical planner, and that two reward-coefficient variants of it did not produce a statistically significant change in coverage at three seeds. Taken together, the entropy correction is at most a stabilizing measure, not a demonstrated route to a competent policy.

The practical point is about method. Entropy is cheap to log per iteration, and three replicate runs per configuration cost a few hours here. Had we done that before writing the diagnosis, we would not have published a causal account of a pattern that does not replicate.

Infrastructure faults. Two distinct faults occurred during this project and are easy to conflate. First, a native segmentation fault in `torch.optim.Adam.__init__`; an earlier draft attributed it to a local installation fault, but it was a TorchDynamo compilation hook failing inside Ray’s actor threads and was resolved by disabling TorchDynamo (`TORCHDYNAMO_DISABLE=1` together with the in-process setting). Second, a later training run stopped at iteration 34 of 45 when a host Wi-Fi interruption severed Ray’s control plane, which had bound to the host’s dynamic address; binding Ray to the loopback interface and capping its object store in-process resolved it. Neither affects the six runs here.

6 Limitations

Three runs per arm is small: the smallest attainable p is 0.10, so no comparison here can reach conventional significance, and we report the values instead. The comparison bundles five changes and cannot attribute an effect to any. Fifteen iterations is short, and entropy might diverge later; the original claim concerned iteration 15. We tested one environment and one architecture. The simulator’s timing is not deterministic, so the runs differ from one another for reasons we do not control, which is part of what a replicate measures. We did not log a task metric or keep the checkpoints, so we make no claim about coverage. The original run’s logs are unavailable, so the failure to reproduce it cannot be traced to a cause.

7 Conclusion

A diagnosis of entropy collapse from default RLlib hyperparameters, formed from one unreplicated run, was not reproduced in three runs of the same configuration: entropy at iteration 15 stayed between 1.50 and 1.87. The corrected configuration raised entropy modestly in all three runs without reaching significance at the attainable sample size, and we did not measure a task benefit. We report the failure to reproduce, the replicated entropy trajectories and the scripts that produced them, in the hope that the baseline is more useful than the original diagnosis was.

Data and code availability. Per-iteration logs for all six runs, the analysis summary and the scripts are in the project repository, under `checkpoints/paper2_ab/` and `scripts/` (files `paper2_ab_entropy.sh` and `paper2_ab_analyze.py`), archived at <https://doi.org/10.5281/zenodo.22758286>.

References

- [1] Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, pages 3207–3214, 2018.
- [2] Jung In Kim, Young Jae Lee, Jongkook Heo, Jinhyeok Park, Jaehoon Kim, Sae Rin Lim, Jinyong Jeong, and Seoung Bum Kim. Sample-efficient multi-agent reinforcement learning with masked reconstruction. *PLoS ONE*, 18(9):e0291545, 2023.
- [3] Harsh Pandhe. Diagnosing a low-displacement failure mode in MAPPO for multi-robot area coverage. SSRN preprint, abstract id 7556699 (under review); code and data: <https://doi.org/10.5281/zenodo.22758286>, 2026.
- [4] Eduardo Sebastian, Thai Duong, Nikolay Atanasov, Eduardo Montijano, and Carlos Sagues. Physics-informed multi-agent reinforcement learning for distributed multi-robot problems, 2024.
- [5] Haojie Shi, Tingguang Li, Qingxu Zhu, Jiapeng Sheng, Lei Han, and Max Q.-H. Meng. An efficient model-based approach on learning agile motor skills without reinforcement, 2024. Accepted to ICRA 2024.

- [6] Chao Yu, Akash Velu, Eugene Vitsitsky, Jiaxuan Gao, Yu Wang, Alexandre Bayen, and Yi Wu. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.
- [7] Yuang Zhang, Yunlong Song, Zhihao He, Zelin Ni, Kangyu Wang, Tianchi Liu, Yu Hu, Feng Yu, Danping Zou, and Weiyao Lin. Asymmetric physics enables efficient learning in quadrupedal robot swarms, 2026.